



# MANIPAL

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**MANIPAL SCHOOL OF INFORMATION SCIENCES**

**(A Constituent unit of MAHE, Manipal)**

## **Cryptology – 5\_Lab**

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**MANIPAL SCHOOL OF INFORMATION SCIENCES**

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## Write Python code to encrypt/decrypt

### -shift cipher

```
def encrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isupper():
            result += chr((ord(ch) - 65 + shift) % 26 + 65)
        elif ch.islower():
            result += chr((ord(ch) - 97 + shift) % 26 + 97)
        else:
            result += ch
    return result

def decrypt(text, shift):
    result = ""
    for ch in text:
        if ch.isupper():
            result += chr((ord(ch) - 65 - shift) % 26 + 65)
        elif ch.islower():
            result += chr((ord(ch) - 97 - shift) % 26 + 97)
        else:
            result += ch
    return result

text = input("Enter text: ")
shift = int(input("Enter shift value: "))
encrypted = encrypt(text, shift)
print("Encrypted text:", encrypted)
decrypted = decrypt(encrypted, shift)
print("Decrypted text:", decrypted)
```

```
● haris@MacBook-M1 5_Lab % python3 1.py
Enter text: Something
Enter shift value: 6
Encrypted text: Yuskznotm
Decrypted text: Something
○ haris@MacBook-M1 5_Lab %
```

### -ceaser cipher

```
def caesar_cipher(text, shift):
    result = ""

    for character in text:
        if character.isalpha():
            # Choose the starting letter: A or a
            if character.isupper():
                start = ord('A')
            else:
                start = ord('a')
            result += chr((ord(character) - start + shift) % 26 + start)
```

```

        # Shift the character
        new_character = chr(
            (ord(character) - start + shift) % 26 + start
        )

        result += new_character
    else:
        # Keep spaces and punctuation unchanged
        result += character

    return result

# Message to encrypt
message = "Hello World!"

# Shift value
shift = 3

# Encrypt the message
encrypted_message = caesar_cipher(message, shift)
print("Encrypted:", encrypted_message)

# Decrypt the message by using a negative shift
decrypted_message = caesar_cipher(encrypted_message, -shift)
print("Decrypted:", decrypted_message)

```

```

● haris@MacBook-M1 5_Lab % python3 2.py
  Encrypted: Khoor Zruog!
  Decrypted: Hello World!
○ haris@MacBook-M1 5_Lab % █

```

## -ROT 13

```

def rot13(text):
    result = ""

    for character in text:
        if character.isalpha():
            if character.isupper():
                start = ord('A')
            else:
                start = ord('a')

            # Shift the letter by 13 positions
            new_character = chr(
                (ord(character) - start + 13) % 26 + start
            )
            result += new_character
        else:
            result += character

    return result

```

```

    )

    result += new_character
else:
    # Keep spaces and punctuation unchanged
    result += character

return result

message = "Hello World!"

# Encrypt using ROT13
encrypted = rot13(message)
print("Encrypted:", encrypted)

# Decrypt using ROT13 again
decrypted = rot13(encrypted)
print("Decrypted:", decrypted)

```

```

● haris@MacBook-M1 5_Lab % python3 3.py
  Encrypted: Uryyb Jbeyq!
  Decrypted: Hello World!
○ haris@MacBook-M1 5_Lab %

```

## -Atbash cipher

```

def atbash_cipher(text):
    result = ""

    for character in text:
        if character.isalpha():
            if character.isupper():
                # Convert A to Z, B to Y, and so on
                new_character = chr(ord('Z') - (ord(character) - ord('A')))
            else:
                # Convert a to z, b to y, and so on
                new_character = chr(ord('z') - (ord(character) - ord('a')))

            result += new_character
        else:
            # Keep spaces and punctuation unchanged
            result += character

    return result

message = "Hello World!"

```

```

encrypted = atbash_cipher(message)
print("Encrypted:", encrypted)

# Applying Atbash again decrypts the message
decrypted = atbash_cipher(encrypted)
print("Decrypted:", decrypted)

```

```

● haris@MacBook-M1 5_Lab % python3 4.py
Encrypted: Svoool Dliow!
Decrypted: Hello World!
○ haris@MacBook-M1 5_Lab % █

```

### -Affine cipher

```

def affine_encrypt(text, a, b):
    result = ""

    for character in text:
        if character.isalpha():
            if character.isupper():
                start = ord('A')
            else:
                start = ord('a')

            # Convert the letter to a number from 0 to 25
            x = ord(character) - start

            # Apply the encryption formula
            encrypted_number = (a * x + b) % 26

            # Convert the number back to a letter
            encrypted_character = chr(encrypted_number + start)
            result += encrypted_character
        else:
            # Keep spaces and punctuation unchanged
            result += character

    return result

def affine_decrypt(text, a, b):
    result = ""

    # Find the modular inverse of a
    a_inverse = pow(a, -1, 26)

    for character in text:
        if character.isalpha():
            if character.isupper():

```

```

        start = ord('A')
    else:
        start = ord('a')

    # Convert the letter to a number from 0 to 25
    y = ord(character) - start

    # Apply the decryption formula
    decrypted_number = (a_inverse * (y - b)) % 26

    # Convert the number back to a letter
    decrypted_character = chr(decrypted_number + start)
    result += decrypted_character
else:
    # Keep spaces and punctuation unchanged
    result += character

return result

message = "Hello World!"

a = 5
b = 8

encrypted = affine_encrypt(message, a, b)
print("Encrypted:", encrypted)

decrypted = affine_decrypt(encrypted, a, b)
print("Decrypted:", decrypted)

```

```

● haris@MacBook-M1 5_Lab % python3 5.py
  Encrypted: Rclla 0aplx!
  Decrypted: Hello World!
○ haris@MacBook-M1 5_Lab %

```